MA104

Discrete Mathematics

Chapter 01

The Foundations: Logic and Proofs

Full Chapter Notes, Worked examples,
reference tables, and practice problems.

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Primary reference: K. H. Rosen, *Discrete Mathematics and Its Applications*, 8th ed.

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1 Course and chapter overview

1.1 Basic course information

Item	Value
Course code	MA104
Course name	Discrete Mathematics
Level	1st Year / B.Sc.
Credit	3 credits
Textbook	K. H. Rosen, *Discrete Mathematics and Its Applications*

1.2 Course objectives

- Learn how to think mathematically.
- Grasp the basic logical and reasoning mechanisms of mathematical thought.
- Acquire logic and proof as the basis for abstract thinking.
- Improve problem-solving skills.
- Grasp the basic elements of induction, recursion, combinatorics, and discrete structures.

1.3 Discrete mathematics is a gateway course

The topics in this course reappear in almost everything that follows:

- **Computer science:** computer architecture, data structures, algorithms, programming languages, compilers, computer security, databases, artificial intelligence, networking, graphics, game design, theory of computation.
- **Mathematics:** logic, set theory, probability, number theory, abstract algebra, combinatorics, graph theory, game theory, network optimisation.
- **Other disciplines:** philosophy, economics, and linguistics all borrow this machinery.

1.4 Course syllabus

1. The Foundations: Logic and Proofs — this chapter
2. Basic Structures: Sets, Functions, Sequences, and Sums
3. Algorithms
4. Induction and Recursion
5. Graphs
6. Trees

1.5 What is inside Chapter 1

1. Introduction to propositional logic
2. Compound propositions
3. Applications of propositional logic
4. Propositional equivalences
5. Predicates and quantifiers
6. Arguments and rules of inference
7. Proof techniques

> **The one-line summary.** This chapter is not a list of symbols. It teaches you to **write an argument that nobody can poke a hole in** — and the symbols are only the grammar of that argument.

2 Introduction to propositional logic

2.1 What is logic?

Logic is the discipline that deals with the methods of reasoning. On an elementary level, logic provides rules and techniques for determining whether a given argument is valid. Logical reasoning is used in mathematics to prove theorems, and in computer science to reason about programs, circuits, and specifications.

2.2 Propositions

Definition. A **proposition** (or **statement**) is a declarative sentence that is either **true** or **false**, but not both. The area of logic that deals with propositions is called **propositional logic**.

The truth value of a proposition is **true**, written T, if it is a true proposition, and **false**, written F, if it is a false proposition. We use lowercase letters $p, q, r, s, \ldots$ as **propositional variables** standing for propositions.

Example 1. Which of these are propositions, and what is the truth value of each?

Sentence	Proposition?	Truth value
$2 + 3 = 5$	yes	**T**
$5 - 2 = 1$	yes	**F**
Today is Friday.	yes	depends on the day
$x + 3 = 7$, for $x = 4$	yes	**T**
Cairo is the capital of Egypt.	yes	**T**
What time is it?	no — a question	—
Read this carefully.	no — a command	—
$x + 3 = 7$	no — x is free	—

Watch out. "$x + 3 = 7$" is **not** a proposition: until x is given a value we cannot say whether it is true or false. Sentences like this are **predicates**, and we return to them in Section 7. Questions, commands, and exclamations are never propositions.

Exercise set 1

1.1 Which of these sentences are propositions? For those that are, give the truth value.

(a) Beni-Suef is a city in Egypt.
(b) $7 + 5 = 11$.
(c) Please close the door.
(d) $x - 4 = 9$.
(e) $x - 4 = 9$ for $x = 13$.
(f) What is the time?

1.2 Explain why "This sentence is false" is not a proposition, even though it is a declarative sentence.

1.3 The sentence "Today is a holiday" is a proposition, but "x is a holiday" is not. Explain the difference in one sentence.

1.4 Give the truth value of each:

- **(a)** $2^5 = 32$.

- **(b)** Every month has 30 days.
- **(c)** Alexandria is the capital of Egypt.

1.5 Write three propositions of your own: one that is true, one that is false, and one whose truth value you cannot decide without extra information.

3 Compound propositions

Definition. **Compound propositions** are formed from existing propositions using **logical operators** (connectives).

Definition. A **truth table** is a table that gives the truth value of a compound proposition for every combination of truth values of its component propositions. A compound proposition built from n propositional variables needs 2^n rows.

3.1 Negation

Definition. Let p be a proposition. The **negation** of p, written $\neg p$, is the statement "it is not the case that p". The proposition $\neg p$ is read "not p"; it is true exactly when p is false.

Other notations you may see for $\neg p$ are $\overline{p}$, $\sim p$, p', and $!p$ (the last one in programming languages).

p	$\neg p$
T	F
F	T

Example 2. Find the negation of the proposition p: "Cairo is the capital of Egypt."

The negation is $\neg p$: "It is not the case that Cairo is the capital of Egypt", which is more simply expressed as $\neg p$: "Cairo is **not** the capital of Egypt."

3.2 The logical connectives

Connective	Symbol	Name	Read as	True exactly when
Negation	$\neg p$	NOT	not p	p is false
Conjunction	$p \wedge q$	AND	p and q	both p and q are true
Disjunction	$p \vee q$	OR (inclusive)	p or q	at least one of p, q is true
Exclusive or	$p \oplus q$	XOR	p or q but not both	exactly one of p, q is true
Conditional	$p \to q$	implication	if p then q	p is false, or q is true
Biconditional	$p \leftrightarrow q$	equivalence	p if and only if q	p and q have the same truth value

3.2.1 Conjunction, disjunction, exclusive or

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$
T	T	T	T	F

p	q	$p \wedge q$	$p \vee q$	$p \oplus q$
T	F	F	T	T
F	T	F	T	T
F	F	F	F	F

Example 3. Let p: "Today is Friday." and q: "It is raining today." Then

- $p \wedge q$: "Today is Friday **and** it is raining today."
- $p \vee q$: "Today is Friday **or** it is raining today (or both)."
- $p \oplus q$: "Today is Friday **or** it is raining today, **but not both**."

A natural XOR: let p: "They are parents." and q: "They are children." Then $p \oplus q$: "They are parents or children but not both."

Watch out. English "or" is ambiguous. "Students who have taken calculus **or** computer science may enrol" is inclusive ($\vee$). "The menu comes with soup **or** salad" is exclusive ($\oplus$). In mathematics, unqualified "or" always means the **inclusive** $\vee$.

Exercise set 2

2.1 Write the negation of each proposition in plain English:
- **(a)** "Today is Thursday."
- **(b)** "There is no niversity in Beni-Suef."
- **(c)** "$2 + 1 = 3$."
- **(d)** "The summer in Egypt is hot and sunny."

2.2 Let p be "It is below freezing" and q be "It is snowing". Write each sentence in symbols:
- **(a)** It is below freezing and snowing.
- **(b)** It is below freezing but not snowing.
- **(c)** It is neither below freezing nor snowing.
- **(d)** It is below freezing or snowing, or both.

2.3 Construct the truth table of $(p \vee q) \wedge \neg(p \wedge q)$. Which connective does it reproduce?

2.4 Construct the truth tables of $\neg(p \wedge q)$ and of $\neg p \vee \neg q$ side by side. What do you notice? (You are rediscovering a law we name in Section 6.)

2.5 Is the "or" inclusive or exclusive in each sentence? (a) "The prize is a laptop or a tablet." (b) "Applicants must know Java or Python." (c) "She was born in 2003 or 2004."

2.6 Simplify, in words and in symbols: "It is not the case that Ahmed is not enrolled."

2.7 How many rows does the truth table of a compound proposition have if it is built from four propositional variables? From n variables?

3.2.2 The conditional statement

Definition. Let p and q be propositions. The **conditional statement** $p \to q$ is the proposition "if p, then q". It is **false** only when p is true and q is false, and **true** otherwise. Here p is the **hypothesis** (antecedent, premise) and q is the **conclusion** (consequent).

p	q	$p \to q$	$p \leftrightarrow q$
T	T	T	T

p	q	$p \to q$	$p \leftrightarrow q$
T	F	F	F
F	T	T	F
F	F	T	T

The conditional $p \to q$ is expressed in English in many ways. All of the following mean $p \to q$:

Phrasing	Phrasing
if p, then q	p implies q
if p, q	p only if q
q if p	q whenever p
q when p	q follows from p
p is sufficient for q	q is necessary for p
a necessary condition for p is q	a sufficient condition for q is p

Watch out. "p **only if** q" is $p \to q$, **not** $q \to p$. This is the single most common translation mistake in the chapter. Read it as: p cannot happen unless q does, so the truth of p forces the truth of q.

Note. If p is false, $p \to q$ is true no matter what q says — the implication is **vacuously true**. "If the moon is made of cheese, then $1 = 2$" is a true statement.

3.2.3 Converse, contrapositive, inverse

Definition. From the conditional $p \to q$ we form three related statements: the **converse** $q \to p$, the **inverse** $\neg p \to \neg q$, and the **contrapositive** $\neg q \to \neg p$.

p	q	$p \to q$	converse $q \to p$	inverse $\neg p \to \neg q$	contrapositive $\neg q \to \neg p$
T	T	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	F	T	T	T	T

Read the table. The contrapositive column is **identical** to the $p \to q$ column: a conditional and its contrapositive are logically equivalent. The converse and the inverse are equivalent to each other, but **not** to the original. This one fact is the engine of proof by contraposition (Section 9.5).

3.2.4 The biconditional

Definition. The **biconditional** $p \leftrightarrow q$ is the proposition "p if and only if q" (abbreviated "p iff q"). It is true exactly when p and q have the same truth value. Note that $p \leftrightarrow q$ is the same as $(p \to q) \wedge (q \to p)$.

3.3 Precedence of logical operators

To cut down on parentheses we fix a precedence order; operators higher in the table bind more tightly.

Operator	Precedence
$\neg$	1 (highest)
$\wedge$	2
$\vee$	3
$\to$	4
$\leftrightarrow$	5 (lowest)

So $\neg p \wedge q$ means $(\neg p) \wedge q$, and $p \vee q \to r$ means $(p \vee q) \to r$.

> **Note.** Precedence saves ink, not clarity. When you write for someone else — a grader, a colleague, a compiler — put the parentheses in anyway.

3.4 Building truth tables for compound propositions

The method never changes: list the 2^n rows of the variables, then add one column per sub-expression, innermost first, until you reach the whole formula.

Example 4. Construct the truth table of $(p \vee \neg q) \to (p \wedge q)$.

p	q	$\neg q$	$p \vee \neg q$	$p \wedge q$	$(p \vee \neg q) \to (p \wedge q)$
T	T	F	T	T	T
T	F	T	T	F	F
F	T	F	F	F	T
F	F	T	T	F	F

Example 5. Construct the truth table of $(p \wedge \neg q) \to r$. Three variables, so $2^3 = 8$ rows.

p	q	r	$\neg q$	$p \wedge \neg q$	$(p \wedge \neg q) \to r$
T	T	T	F	F	T
T	T	F	F	F	T
T	F	T	T	T	T
T	F	F	T	T	F
F	T	T	F	F	T
F	T	F	F	F	T
F	F	T	T	F	T
F	F	F	T	F	T

Only one row makes the statement false — the row where the hypothesis $p \wedge \neg q$ holds but the conclusion r fails.

3.5 Logic and bit operations

Computers represent information using **bits**. A bit is a symbol with two possible values, 0 (zero) and 1 (one). We identify 1 with T and 0 with F, which turns every logical connective into a bit operation. In programming languages these are usually written `OR`, `AND`, `XOR` for $\vee$, $\wedge$, $\oplus$.

x	y	$x \vee y$	$x \wedge y$	$x \oplus y$
0	0	0	0	0
0	1	1	0	1
1	0	1	0	1
1	1	1	1	0

Definition. A **bit string** is a sequence of zero or more bits; its **length** is the number of bits in it. The **bitwise** OR, AND, and XOR of two strings of the same length are obtained by applying the operator to corresponding bits.

Example 6. Find the bitwise OR, AND, and XOR of the bit strings `01 1011 0110` and `11 0001 1101`.

	`01 1011 0110`
	`11 0001 1101`
bitwise OR	`11 1011 1111`
bitwise AND	`01 0001 0100`
bitwise XOR	`10 1010 1011`

Exercise set 3

3.1 Let p be "It is below freezing" and q be "It is snowing". Write each sentence in symbols: (a) If it is below freezing, it is also snowing. (b) It is below freezing if and only if it is snowing. (c) It is snowing whenever it is below freezing. (d) It is below freezing only if it is snowing.

3.2 Construct the truth table of $\neg p \vee q$ and compare it with the truth table of $p \to q$. What do you conclude?

3.3 Construct the truth table of $(p \to q) \leftrightarrow (\neg q \to \neg p)$ and say what it shows.

3.4 Write the converse, the inverse, and the contrapositive of "If it snows today, then I will ski tomorrow." Which of the three is equivalent to the original?

3.5 What is the truth value of "If $1 + 1 = 3$, then unicorns exist"? Justify your answer from the truth table of $\to$, not from biology.

3.6 Insert all the parentheses that the precedence rules leave implicit in $\neg p \wedge q \to r \vee \neg s$.

3.7 Find the bitwise OR, AND, and XOR of the bit strings `1111 0000` and `1010 1010`.

4 Applications of propositional logic

Propositional logic earns its keep in five places: translating English, writing system specifications, Boolean searches, logic puzzles, and logic circuits.

4.1 Translating English sentences

English — like every human language — is ambiguous. Translating a sentence into propositional variables and connectives removes the ambiguity, at the cost of some nuance. The recipe:

1. Find the atomic statements and name them $p, q, r, \ldots$ (always state them **positively**).
2. Find the connective words: *and*, *or*, *not*, *if … then*, *only if*, *unless*, *if and only if*.
3. Assemble, and put the parentheses in.

Example 7. "You can access the Internet from campus only if you are a computer science major or you are not a freshman."

Let

- p: "You can access the Internet from campus."
- q: "You are a computer science major."
- r: "You are a freshman."

The sentence has the shape "(p) **only if** (q or not r)", and "a only if b" is $a \to b$. Therefore the translation is

$$p \to (q \vee \neg r).$$

Example 8. "The automated reply cannot be sent when the file system is full."

Let p: "The automated reply can be sent." and q: "The file system is full." The shape is "(not p) **when** (q)", and "b when a" is $a \to b$. Therefore

$$q \to \neg p.$$

Note. Two English words that flip the arrow: **"only if"** puts the arrow forward (p only if q is $p \to q$), **"when / whenever / if"** puts the clause it introduces on the left (q when p is $p \to q$). "p unless q" is $\neg q \to p$.

4.2 System specifications

Specifications for software and hardware are written as compound propositions so that they can be checked for **consistency**: a list of specifications is consistent if there is an assignment of truth values making all of them true simultaneously.

Example 9. Are these specifications consistent?

- "The diagnostic message is stored in the buffer or it is retransmitted."
- "The diagnostic message is not stored in the buffer."
- "If the diagnostic message is stored in the buffer, then it is retransmitted."

With p: "stored in the buffer" and q: "retransmitted", the specifications are $p \vee q$, $\neg p$, and $p \to q$. Setting $p = F$ and $q = T$ makes all three true, so the specifications are consistent. Adding "the message is not retransmitted" ($\neg q$) would make them inconsistent — no assignment satisfies all four.

4.3 Boolean searches

Search engines use $\wedge$, $\vee$, and $\neg$ to combine keywords: `NEW AND MEXICO AND UNIVERSITIES` narrows, `(NEW AND MEXICO OR ARIZONA) AND UNIVERSITIES` widens, and `MEXICO AND UNIVERSITIES NOT NEW` excludes.

4.4 Logic puzzles

Puzzles that can be solved by reasoning about truth values are **logic puzzles**. The technique is always the same: name the atomic claims, write each statement of the puzzle as a compound proposition, then eliminate the assignments that make some statement false.

Example 10. On an island each inhabitant is either a **knight** (always tells the truth) or a **knave** (always lies). You meet A and B. A says "B is a knight", B says "the two of us are of opposite types". What are they?

Let p: "A is a knight", q: "B is a knight". If p is true, A's statement is true, so q is true; then B is a knight and B's statement must be true, so A and B are of opposite types — contradicting that both are knights. So p is false: A is a knave, A's statement is a lie, hence q is false and B is a knave as well. Check: B, a knave, claims they are of opposite types — false, as required. **Both are knaves.**

4.5 Logic circuits

Definition. A **logic circuit** (digital circuit) receives input signals $p_1, p_2, ..., p_n$, each a bit (0 = off, 1 = on), and produces output signals, each a bit. We restrict attention to circuits with a single output.

Complicated digital circuits are built from three basic gates:

Inverter	AND gate	OR gate
NOT	AND	OR
$p \mapsto \neg p$	$p, q \mapsto p \wedge q$	$p, q \mapsto p \vee q$

Example 11. Determine the output of the circuit that feeds p and $\neg q$ into an AND gate, and feeds the result together with $\neg r$ into an OR gate.

$$p \text{ — } \boxed{\text{AND}} \text{ — } (p \wedge \neg q) \text{ — } \boxed{\text{OR}} \text{ — } (p \wedge \neg q) \vee \neg r$$

Reading the circuit gate by gate: the inverter on q produces $\neg q$; the AND gate produces $p \wedge \neg q$; the inverter on r produces $\neg r$; the OR gate produces the output

$$(p \wedge \neg q) \vee \neg r.$$

Example 12. Build a digital circuit that produces the output

$$(p \vee \neg r) \wedge (\neg p \vee (q \vee \neg r)).$$

Work outside-in, and build each parenthesis as one sub-circuit:

1. Invert r to get $\neg r$; invert p to get $\neg p$.
2. OR gate on p and $\neg r$ $\to$ $(p \vee \neg r)$.
3. OR gate on q and $\neg r$ $\to$ $(q \vee \neg r)$.
4. OR gate on $\neg p$ and the output of step 3 $\to$ $(\neg p \vee (q \vee \neg r))$.
5. AND gate on the outputs of steps 2 and 4 $\to$ the required output.

Stage	Gate	Inputs	Output signal
1	NOT	r	$\neg r$

Stage	Gate	Inputs	Output signal
1	NOT	p	$\neg p$
2	OR	$p, \neg r$	$p \lor \neg r$
3	OR	$q, \neg r$	$q \lor \neg r$
4	OR	$\neg p, (q \lor \neg r)$	$\neg p \lor (q \lor \neg r)$
5	AND	stage 2, stage 4	$(p \lor \neg r) \land (\neg p \lor (q \lor \neg r))$

Exercise set 4

4.1 Translate into logic, stating your propositional variables: "You can ride the roller coaster if you are under 4 feet tall, unless you are older than 16 years old."

4.2 Translate into logic: "Access is granted only if the user is an administrator or the file is public."

4.3 Are these system specifications consistent? $p \leftrightarrow q$ ("the system is in multiuser state if and only if it is operating normally"), $q \to r$ ("if it is operating normally, the kernel is functioning"), $\neg r \lor s$ ("the kernel is not functioning or the system is in interrupt mode"), $\neg p \to s$ ("if the system is not in multiuser state, then it is in interrupt mode"), and $\neg s$ ("the system is not in interrupt mode").

4.4 On the island of knights and knaves you meet A and B. A says "I am a knave or B is a knight." What are A and B?

4.5 Give the output of the circuit that feeds p and $\neg q$ into an AND gate, feeds $\neg p$ and r into a second AND gate, and feeds both results into an OR gate. What is the output bit when $p = 1$, $q = 1$, $r = 0$?

5 Classifying compound propositions

Definition. A compound proposition that is **always true**, no matter the truth values of its variables, is a **tautology**. One that is **always false** is a **contradiction**. One that is neither — sometimes true and sometimes false — is a **contingency**.

p	$\neg p$	$p \lor \neg p$ (tautology)	$p \land \neg p$ (contradiction)
T	F	T	F
F	T	T	F

Example 13. Show that $(p \land q) \to p$ is a tautology using a truth table.

p	q	$p \land q$	$(p \land q) \to p$
T	T	T	T
T	F	F	T

p	q	$p \wedge q$	$(p \wedge q) \to p$
F	T	F	T
F	F	F	T

The last column is all T, so the statement is a tautology.

Exercise set 5

5.1 Classify each as a tautology, a contradiction, or a contingency: (a) $p \vee \neg p$; (b) $p \wedge \neg p$; (c) $(p \to q) \wedge p$; (d) $((p \to q) \wedge q) \to p$.

5.2 Show by truth table that $(p \to q) \vee (q \to p)$ is a tautology. Explain in words why no row can make it false.

5.3 Show that $(p \wedge \neg q) \wedge (\neg p \vee q)$ is a contradiction.

5.4 Is $\neg(p \leftrightarrow q) \leftrightarrow (p \leftrightarrow \neg q)$ a tautology? Justify.

6 Propositional equivalences

Definition. Compound propositions p and q that have the same truth values in **all** possible cases are **logically equivalent**, written $p \equiv q$ (or $p \Leftrightarrow q$). Equivalently: $p \equiv q$ exactly when $p \leftrightarrow q$ is a tautology.

Note. $\equiv$ is not a connective — it is a claim **about** two formulas. $p \leftrightarrow q$ is a proposition; $p \equiv q$ is a statement that we have checked every row.

6.1 Two equivalences worth memorising

Example 14. Show that $p \to q \equiv \neg p \vee q$.

p	q	$p \to q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Columns 3 and 5 agree in every row, so the two are equivalent. This is the rewriting rule that turns every implication into a disjunction — indispensable in proofs by resolution.

Example 15. Show De Morgan's law $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F

p	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
F	F	F	T	T	T	T

6.2 Table of logical equivalences

Equivalence	Name
$p \wedge T \equiv p$ $p \vee F \equiv p$	Identity laws
$p \vee T \equiv T$ $p \wedge F \equiv F$	Domination laws
$p \vee p \equiv p$ $p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$	Double negation law
$p \vee q \equiv q \vee p$ $p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$ $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \wedge q) \equiv \neg p \vee \neg q$ $\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan's laws
$p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv T$ $p \wedge \neg p \equiv F$	Negation laws

6.2.1 Equivalences involving conditionals

Equivalence	Comment
$p \to q \equiv \neg p \vee q$	the rewriting rule
$p \to q \equiv \neg q \to \neg p$	contrapositive
$\neg(p \to q) \equiv p \wedge \neg q$	how to negate an implication
$p \vee q \equiv \neg p \to q$	
$p \wedge q \equiv \neg(p \to \neg q)$	
$(p \to q) \wedge (p \to r) \equiv p \to (q \wedge r)$	
$(p \to r) \wedge (q \to r) \equiv (p \vee q) \to r$	
$(p \to q) \vee (p \to r) \equiv p \to (q \vee r)$	
$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r$	

6.2.2 Equivalences involving biconditionals

$p \leftrightarrow q \equiv (p \to q) \wedge (q \to p)$
$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$

6.3 Proving equivalences by rewriting

Truth tables always work, but they grow as 2^n. The alternative is to **rewrite** one side into the other, naming the law used at each step. This is how you will be expected to work on an exam.

Example 16. Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent.

$\neg(p \vee (\neg p \wedge q))$	given
$\equiv \neg p \wedge \neg(\neg p \wedge q)$	De Morgan's second law
$\equiv \neg p \wedge (\neg(\neg p) \vee \neg q)$	De Morgan's first law
$\equiv \neg p \wedge (p \vee \neg q)$	double negation law
$\equiv (\neg p \wedge p) \vee (\neg p \wedge \neg q)$	distributive law
$\equiv F \vee (\neg p \wedge \neg q)$	negation law, $\neg p \wedge p \equiv F$
$\equiv (\neg p \wedge \neg q) \vee F$	commutative law
$\equiv \neg p \wedge \neg q$	identity law ■

How to write these. One rewriting per line, the name of the law on the right, and never two laws in one step. A grader cannot give credit for a jump they cannot follow — and a step you cannot name is usually a step that is wrong.

Exercise set 6

6.1 Verify De Morgan's law $\neg(p \wedge q) \equiv \neg p \vee \neg q$ with a truth table.
6.2 Use the laws of logic — no truth table — to show that $(p \wedge q) \to p$ is a tautology.
6.3 Simplify $(p \wedge q) \vee (p \wedge \neg q)$ to a single variable, naming the law at each step.
6.4 Prove the **exportation law** $p \to (q \to r) \equiv (p \wedge q) \to r$ by rewriting.
6.5 Negate and simplify until no $\neg$ stands in front of a bracket: $\neg(p \to (q \wedge r))$.
6.6 Show that $\neg(p \oplus q) \equiv p \leftrightarrow q$.

7 Predicates and quantifiers

7.1 Predicates

Consider the sentence "x is greater than 3". It has two parts: the **subject** x and the **predicate** "is greater than 3", which states a property the subject may have. We write this sentence as $P(x)$, where P denotes the predicate and x is the variable.

Definition. $P(x)$ is the value of the **propositional function** P at x. Once a value has been assigned to x, the statement $P(x)$ becomes a proposition and has a truth value.

Example 17. Let $P(x)$ be "$x > 3$". What are the truth values of $P(4)$ and $P(2)$?

Setting $x = 4$ gives "$4 > 3$", which is **true**. Setting $x = 2$ gives "$2 > 3$", which is **false**.

Example 18. Let $A(x)$ be "computer x is under attack". Suppose only `CS2` and `MATH1` are under attack. Then $A(\text{CS1})$ is false, $A(\text{CS2})$ is true, $A(\text{MATH1})$ is true.

Example 19. Predicates can take several variables. Let $Q(x, y)$ be "$x = y + 3$". Then $Q(1, 2)$ is "$1 = 5$", false, and $Q(3, 0)$ is "$3 = 3$", true. A statement of the form $Q(x_1, ..., x_n)$ is the value of an ***n*-place predicate**.

7.2 Quantifiers

Quantification expresses the extent to which a predicate is true over a range of elements. The **domain of discourse** (universe) U is the collection of values the variable may take; a quantified statement means nothing until the domain is fixed.

Quantifier	Notation	Read	True when
Universal	$\forall x P(x)$	for all x, $P(x)$	$P(x)$ is true for **every** x in U
Existential	$\exists x P(x)$	there exists x such that $P(x)$	$P(x)$ is true for **at least one** x in U
Uniqueness	$\exists! x P(x)$	there is exactly one x with $P(x)$	$P(x)$ is true for **exactly one** x in U

The uniqueness quantifier is also written $\exists_1 x P(x)$.

Statement	When true	When false
$\forall x P(x)$	$P(x)$ is true for every x	there is an x for which $P(x)$ is false
$\exists x P(x)$	there is an x for which $P(x)$ is true	$P(x)$ is false for every x

Note. A **counterexample** is exactly the object whose existence falsifies $\forall x P(x)$: one x with $\neg P(x)$. To disprove a universal claim you need one; to prove one, a hundred examples are worth nothing.

Example 20. Let $P(x)$ be "$x + 1 > x$". What is the truth value of $\forall x P(x)$ over the real numbers? True — adding one to any real number yields a larger number.

Example 21. Let $Q(x)$ be "$x < 2$". What is the truth value of $\forall x Q(x)$ when the domain is all real numbers? False: $x = 3$ is a counterexample, since $3 < 2$ is false.

Example 22. Let $P(x)$ be "$x^2 > 10$" with domain $\{1, 2, 3, 4\}$. Then $\exists x P(x)$ is true, because $P(4)$ is "$16 > 10$", which is true. Note $\forall x P(x)$ is false here, since $P(1)$ fails.

Example 23. When the domain is finite, say $U = \{x_1, ..., x_n\}$, quantifiers collapse into connectives:

$$\forall x P(x) \equiv P(x_1) \wedge P(x_2) \wedge ... \wedge P(x_n), \quad \exists x P(x) \equiv P(x_1) \vee P(x_2) \vee ... \vee P(x_n).$$

This is why $\forall$ behaves like a long AND and $\exists$ like a long OR.

7.3 Precedence and scope

$\forall$ and $\exists$ have **higher precedence** than all logical operators from propositional calculus. So $\forall x P(x) \vee Q(x)$ means $(\forall x P(x)) \vee Q(x)$, not $\forall x(P(x) \vee Q(x))$.

A variable is **bound** when a quantifier is applied to it, and **free** otherwise. A statement with a free variable is not a proposition. The part of a formula to which a quantifier is applied is its **scope**.

7.4 Translating with quantifiers

Example 24. Express "Every student in this class has studied calculus."

Let $S(x)$: "x is in this class", $P(x)$: "x has studied calculus", with domain all people. Then

$$\forall x(S(x) \to P(x)).$$

If instead the domain is **just the students of this class**, the translation is simply $\forall x P(x)$.

Watch out. $\forall x(S(x) \wedge P(x))$ is a **wrong** translation of "every student in this class has studied calculus" — it says every person in the universe is a student in this class **and** has studied calculus. With $\forall$ use $\to$; with $\exists$ use $\wedge$: "some student in this class has studied calculus" is $\exists x(S(x) \wedge P(x))$.

Example 25. Translate $\forall x(C(x) \vee \exists y(C(y) \wedge F(x,y)))$ into English, where $C(x)$ is "x has a computer", $F(x,y)$ is "x and y are friends", and the domain for both x and y is all students in your school.

"Every student in your school has a computer, or has a friend who has a computer."

Example 26. Translate "Everyone has exactly one best friend" into logic. Let $B(x,y)$ be "y is the best friend of x", domain all people.

$$\forall x\, \exists y\, \forall z(B(x,y) \wedge ((z \neq y) \to \neg B(x,z))).$$

7.5 Nested quantifiers

The **order** of different quantifiers matters. With domain the real numbers and $P(x,y)$ meaning "$x + y = 0$":

Statement	Meaning	Truth value
$\forall x\, \exists y\, P(x,y)$	for every x there is a y with $x + y = 0$	true — take $y = -x$
$\exists y\, \forall x\, P(x,y)$	there is one y that works for every x	false

The rule of thumb. $\forall\exists$ lets the witness depend on the choice; $\exists\forall$ demands one witness that works for everything. Swapping them changes the claim, and usually its truth value. Same-type quantifiers ($\forall\forall$, $\exists\exists$) may be swapped freely.

7.6 Logical equivalences involving quantifiers

Definition. Statements involving predicates and quantifiers are **logically equivalent** if and only if they have the same truth value no matter which predicates are substituted and which domain is used.

$$\forall x(P(x) \wedge Q(x)) \equiv \forall x P(x) \wedge \forall x Q(x)$$

Proof. We show each side implies the other.

($\rightarrow$) Assume $\forall x(P(x) \wedge Q(x))$ is true. Then for every x in the domain $P(x) \wedge Q(x)$ is true, hence $P(x)$ is true for all x and $Q(x)$ is true for all x. So $\forall x P(x)$ is true and $\forall x Q(x)$ is true, i.e. $\forall x P(x) \wedge \forall x Q(x)$ is true.

($\leftarrow$) Assume $\forall x P(x) \wedge \forall x Q(x)$ is true. Then $\forall x P(x)$ is true and $\forall x Q(x)$ is true, so for every x both $P(x)$ and $Q(x)$ hold, hence $P(x) \wedge Q(x)$ holds for every x, i.e. $\forall x(P(x) \wedge Q(x))$ is true. ■

Watch out. $\forall$ distributes over $\wedge$ and $\exists$ distributes over $\vee$ — and **not** the other way round. $\forall x(P(x) \vee Q(x))$ is weaker than $\forall x P(x) \vee \forall x Q(x)$: every integer is even or odd, but it is false that every integer is even or that every integer is odd.

7.7 Negating quantified expressions

De Morgan's laws for quantifiers.

$$\neg \forall x P(x) \equiv \exists x \neg P(x), \qquad \neg \exists x P(x) \equiv \forall x \neg P(x).$$

Example 27. Let $P(x)$ be "x has taken a course in calculus", with domain the students in your class.

$\forall x P(x)$ reads "Every student in your class has taken a course in calculus." Its negation is "There is at least one student in your class who has **not** taken a course in calculus", i.e. $\neg \forall x P(x) \equiv \exists x \neg P(x)$.

$\exists x P(x)$ reads "At least one student in your class has taken a course in calculus." Its negation is "Every student in this class has not taken calculus", i.e. $\neg \exists x P(x) \equiv \forall x \neg P(x)$.

Example 28. What is the negation of $\forall x(x^2 > x)$?

$$\neg \forall x(x^2 > x) \equiv \exists x \neg(x^2 > x) \equiv \exists x(x^2 \leq x).$$

Example 29. What is the negation of $\exists x(x^2 = 2)$?

$$\neg \exists x(x^2 = 2) \equiv \forall x \neg(x^2 = 2) \equiv \forall x(x^2 \neq 2).$$

Mechanical negation. Push the $\neg$ inwards, one symbol at a time: every $\forall$ becomes $\exists$, every $\exists$ becomes $\forall$, $\wedge$ becomes $\vee$, $\vee$ becomes $\wedge$, $P \rightarrow Q$ becomes $P \wedge \neg Q$, and the innermost predicate gets negated ($> \mapsto \leq$, $= \mapsto \neq$). Never leave a $\neg$ sitting in front of a quantifier in a final answer.

Example 30. Negate $\forall x \, \exists y \, (P(x, y) \rightarrow Q(x, y))$ step by step.

$$\neg \forall x \exists y(P \rightarrow Q) \equiv \exists x \neg \exists y(P \rightarrow Q) \equiv \exists x \forall y \neg(P \rightarrow Q) \equiv \exists x \forall y(P \wedge \neg Q).$$

Exercise set 7

7.1 Let $P(x)$ be "$x^2 \geq x$". What is the truth value of $\forall x P(x)$ when the domain is all integers? When it is all real numbers?

7.2 Let $S(x)$ be "x is a student in this class", $M(x)$ be "x has visited Mexico", and $C(x)$ be "x has visited Canada", with domain all people. Translate: (a) every student in this class has visited Mexico; (b) some student in this class has visited Canada; (c) no student in this class has visited Canada.

7.3 Express "there is exactly one student in this class who solved the problem" using the uniqueness quantifier, and then again using only $\forall$ and $\exists$.

7.4 What is the truth value of $\forall x\,\exists y\,(xy = 1)$ when the domain is all real numbers? When it is all **non-zero** real numbers?

7.5 Negate, pushing the $\neg$ all the way in: (a) $\exists x(x^2 = x + 1)$; (b) $\forall x \forall y(x + y = y + x)$; (c) $\forall x(x > 0 \to \exists y(y^2 = x))$.

7.6 Do $\forall x(P(x) \lor Q(x))$ and $\forall x P(x) \lor \forall x Q(x)$ say the same thing? Give a domain and predicates that settle it.

7.7 Translate into English: $\exists x(C(x) \land \forall y(C(y) \to (y = x)))$, where $C(x)$ means "x is in this class and owns a car".

8 Rules of inference

8.1 Valid arguments in propositional logic

Definition. An **argument** in propositional logic is a sequence of propositions. All propositions in the argument except the final one are the **premises** (hypotheses); the final proposition is the **conclusion**. The argument is **valid** if the conjunction of the premises implies the conclusion — that is, if $(p_1 \land p_2 \land \ldots \land p_n) \to q$ is a tautology.

Consider:

$$\begin{array}{ll} p \to q & \textit{"If you have a current password, then you can log onto the network."} \\ p & \textit{"You have a current password."} \\ \hline \therefore\, q & \textit{"Therefore, you can log onto the network."} \end{array}$$

This argument is valid if $((p \to q) \land p) \to q$ is a tautology — and it is:

p	q	$p \to q$ (prem. 1)	p (prem. 2)	$(p \to q) \land p$	q (concl.)	$((p \to q) \land p) \to q$
T	T	T	T	T	T	T
T	F	F	T	F	F	T
F	T	T	F	F	T	T
F	F	T	F	F	F	T

Note. **Valid** is not the same as **true**. A valid argument with a false premise can have a false conclusion; validity only guarantees that the conclusion holds **whenever** all premises do.

8.2 The rules of inference

Truth tables get impractical fast — a proof with ten variables would need 1024 rows. Instead we chain together small arguments already known to be valid.

Rule	Tautology	Name
$p \to q$ p $\therefore\, q$	$((p \to q) \land p) \to q$	Modus ponens

Rule	Tautology	Name
$p \to q$ $\neg q$ $\therefore \neg p$	$((p \to q) \wedge \neg q) \to \neg p$	Modus tollens
$p \to q$ $q \to r$ $\therefore p \to r$	$((p \to q) \wedge (q \to r)) \to (p \to r)$	Hypothetical syllogism
$p \vee q$ $\neg p$ $\therefore q$	$((p \vee q) \wedge \neg p) \to q$	Disjunctive syllogism
p $\therefore p \vee q$	$p \to (p \vee q)$	Addition
$p \wedge q$ $\therefore p$	$(p \wedge q) \to p$	Simplification
p q $\therefore p \wedge q$	$((p) \wedge (q)) \to (p \wedge q)$	Conjunction
$p \vee q$ $\neg p \vee r$ $\therefore q \vee r$	$((p \vee q) \wedge (\neg p \vee r)) \to (q \vee r)$	Resolution

Example 31. Use a truth table to show that the hypotheses $p \vee q$ and $\neg p \vee r$ lead to the conclusion $q \vee r$ (this is resolution).

p	q	r	$p \vee q$	$\neg p$	$\neg p \vee r$	$(p \vee q) \wedge (\neg p \vee r)$	$q \vee r$
T	T	T	T	F	T	T	T
T	T	F	T	F	F	F	T
T	F	T	T	F	T	T	T
T	F	F	T	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	F

In every row where the conjunction of the premises is T (rows 1, 3, 5, 6), the conclusion $q \vee r$ is also T. The argument is valid. Equivalently, the implication from premises to conclusion is a tautology.

Example 32. Using the rules of inference, show that the hypotheses $\neg p \wedge q$, $r \to p$, $\neg r \to s$, $s \to t$ lead to the conclusion t.

1.	$\neg p \wedge q$	Premise
2.	$\neg p$	Simplification from (1)
3.	$r \to p$	Premise
4.	$\neg r$	Modus tollens from (2) and (3)

5.	$\neg r \to s$	Premise
6.	s	Modus ponens from (4) and (5)
7.	$s \to t$	Premise
8.	t	Modus ponens from (6) and (7) ■

Example 33. Show that the premises "If you send me an e-mail message, then I will finish writing the program", "If you do not send me an e-mail message, then I will go to sleep early", and "If I go to sleep early, then I will wake up feeling refreshed" lead to the conclusion "If I do not finish writing the program, then I will wake up feeling refreshed."

Name the atoms:

- p: "You send me an e-mail message."
- q: "I will finish writing the program."
- r: "I will go to sleep early."
- s: "I will wake up feeling refreshed."

Premises: $p \to q$, $\neg p \to r$, $r \to s$. Conclusion: $\neg q \to s$.

1.	$p \to q$	Premise 1
2.	$\neg q \to \neg p$	Contrapositive of (1)
3.	$\neg p \to r$	Premise 2
4.	$\neg q \to r$	Hypothetical syllogism from (2) and (3)
5.	$r \to s$	Premise 3
6.	$\neg q \to s$	Hypothetical syllogism from (4) and (5) ■

Example 34. The same argument, done by **resolution** instead — rewrite every implication as a disjunction and cancel complementary literals.

1.	$p \to q$	Premise 1
2.	$\neg p \lor q$	Logical equivalence of (1)
3.	$\neg p \to r$	Premise 2
4.	$p \lor r$	Logical equivalence of (3)
5.	$q \lor r$	Resolution on (2) and (4)
6.	$r \to s$	Premise 3
7.	$\neg r \lor s$	Logical equivalence of (6)
8.	$q \lor s$	Resolution on (5) and (7)
9.	$\neg q \to s$	Logical equivalence of (8) ■

8.3 Rules of inference for quantified statements

Rule	Name and meaning
$\forall x P(x)$ $\therefore P(c)$	**Universal instantiation** — from "all" to a particular c in the domain.
$P(c)$ for an arbitrary c $\therefore \forall x P(x)$	**Universal generalisation** — c must be arbitrary: nothing may have been assumed about it.

Rule	Name and meaning
$\exists x P(x)$ $\therefore P(c)$ for some element c	**Existential instantiation** — name the witness c; the name must be **fresh**.
$P(c)$ for some element c $\therefore \exists x P(x)$	**Existential generalisation** — one witness is enough.

Watch out. Existential instantiation must use a **new** constant. If two premises each say "there exists", you may not give both witnesses the same name — that quietly assumes they are the same object.

Example 35. Show that the premises "A student in this class has not read the book" and "Everyone in this class passed the first exam" imply "Someone who passed the first exam has not read the book."

Let $P(x)$: "x is in this class", $Q(x)$: "x has read the book", $S(x)$: "x passed the first exam".

Premises: $\exists x(P(x) \wedge \neg Q(x))$ and $\forall x(P(x) \to S(x))$. Conclusion: $\exists x(S(x) \wedge \neg Q(x))$.

1.	$\exists x(P(x) \wedge \neg Q(x))$	Premise 1
2.	$P(a) \wedge \neg Q(a)$	Existential instantiation from (1)
3.	$P(a)$	Simplification from (2)
4.	$\neg Q(a)$	Simplification from (2)
5.	$\forall x(P(x) \to S(x))$	Premise 2
6.	$P(a) \to S(a)$	Universal instantiation from (5)
7.	$S(a)$	Modus ponens from (3) and (6)
8.	$S(a) \wedge \neg Q(a)$	Conjunction from (7) and (4)
9.	$\exists x(S(x) \wedge \neg Q(x))$	Existential generalisation from (8) ■

Exercise set 8

8.1 Name the rule of inference used in each argument: (a) "Alice is a mathematics major. Therefore Alice is a mathematics major or a computer science major." (b) "Jerry is a mathematics major and a computer science major. Therefore Jerry is a mathematics major." (c) "If it is rainy, the pool is closed. It is rainy. Therefore the pool is closed." (d) "If I go swimming, I will stay in the sun too long. If I stay in the sun too long, I will sunburn. Therefore if I go swimming, I will sunburn."

8.2 Is this argument valid? "If you do every problem in the book, you will learn discrete mathematics. You learned discrete mathematics. Therefore you did every problem in the book." Name the fallacy if it is not.

8.3 Show that the premises $\neg q$ and $p \to q$ lead to $\neg p$, first by truth table and then by naming a single rule of inference.

8.4 Show that the premises $p \to (q \to r)$, p, and q lead to the conclusion r.

8.5 Show that the premises $p \vee q$, $\neg p \vee r$, and $\neg q \vee r$ lead to the conclusion r using resolution only.

8.6 Explain why the argument "$\exists x P(x)$, $\exists x Q(x)$, therefore $\exists x(P(x) \wedge Q(x))$" is invalid, and give predicates over the integers that make the premises true and the conclusion false.

9 Proof techniques

9.1 Some terminology

Term	Meaning
Theorem	A statement that can be shown to be true (also: fact, result).
Proof	A valid argument that establishes the truth of a theorem.
Axiom	A statement assumed true, used as a basis for proving theorems.
Lemma	A "helping theorem" used in the proof of a larger theorem.
Corollary	A theorem that follows directly and with little work from another.
Conjecture	A statement proposed to be true, not yet proved.

9.2 The definitions you will need

Object	How you are allowed to write it
a is **even**	$a = 2n$ for some integer n $\quad(2 \times$ any integer is even$)$
a is **odd**	$a = 2m + 1$ for some integer m
a is a **perfect square**	$a = n^2$ for some integer n
a is **rational**	$a = \frac{n}{m}$ for integers n, m with no common factor and $m \neq 0$
a **divides** b	$b = ak$ for some integer k; written $a \mid b$

Useful negations: ¬ even = odd, ¬ odd = even, ¬ rational = irrational, ¬ irrational = rational. Also: even $+1$ = odd and odd $+1$ = even.

9.3 The map of proof techniques

Proof techniques for $p \to q$		
Direct proof	**Indirect proof**	
assume p, derive q	**Contraposition**: assume $\neg q$, derive $\neg p$	**Contradiction**: assume $p \wedge \neg q$, derive a falsehood

9.4 Direct proof

Skeleton. To prove $p \to q$ directly:
1. Assume that p is true.
2. Using definitions, axioms, and known theorems, show that q must also be true.
3. Conclude that $p \to q$ is true.

You never consider the case where p is false — the implication is automatically true there.

Example 36. Give a direct proof of the theorem "If n is an odd integer, then n^2 is odd."

Here p is "n is an odd integer" and q is "n^2 is odd".

Proof. Assume p is true: n is odd, so by definition $n = 2m + 1$ for some integer m. Then

$$n^2 = (2m+1)^2 = 4m^2 + 4m + 1 = 2(2m^2 + 2m) + 1.$$

Since $2m^2 + 2m$ is an integer, n^2 has the form $2k + 1$ — even plus one, which is odd. Hence q is true, and therefore $p \to q$ is true. ■

Example 37. Give a direct proof that if m and n are both perfect squares, then mn is a perfect square.

Proof. Assume m and n are perfect squares, so $m = s^2$ and $n = t^2$ for some integers s and t. Then $mn = s^2t^2 = (st)^2$, and st is an integer. Hence mn is a perfect square. ■

9.5 Proof by contraposition

Because $p \to q \equiv \neg q \to \neg p$, proving the contrapositive proves the original. Reach for this when the hypothesis p is hard to use but $\neg q$ is easy to write down algebraically.

Skeleton. To prove $p \to q$ by contraposition:
1. Assume that $\neg q$ is true.
2. Show that $\neg p$ must also be true.
3. Conclude that $\neg q \to \neg p$ is true, hence so is $p \to q$.

Example 38. Prove by contraposition: if n is an integer and $3n + 2$ is odd, then n is odd.

Here p is "$3n + 2$ is odd" and q is "n is odd"; so $\neg q$ is "n is even" and $\neg p$ is "$3n + 2$ is even".

Proof. Assume $\neg q$: n is even, so $n = 2m$ for some integer m. Then

$$3n + 2 = 3(2m) + 2 = 6m + 2 = 2(3m + 1),$$

which is $2 \times$ an integer, hence even. So $3n + 2$ is even, which is $\neg p$. Therefore $\neg q \to \neg p$ is true, and so $p \to q$ is true. ■

9.6 Proof by contradiction

Skeleton — two forms. **A. To prove a bare statement p.** Show that $\neg p \to F$, i.e. that assuming $\neg p$ leads to a false statement (a contradiction such as $r \wedge \neg r$). Since the implication is true and its conclusion is false, $\neg p$ must be false, so p is true.

B. To prove $p \to q$.
1. Assume the negation of the conclusion, $\neg q$, together with p.
2. Show that $(p \wedge \neg q) \to F$.
3. Since $((p \wedge \neg q) \to F) \equiv \neg(p \wedge \neg q) \equiv (p \to q)$, we are done.

Example 39. Give a proof by contradiction of "If $3n + 2$ is odd, then n is odd."

Proof. Assume $p \wedge \neg q$: that $3n + 2$ is odd **and** n is even. Since n is even, $n = 2m$ for an integer m, so

$$3n + 2 = 6m + 2 = 2(3m + 1),$$

which is even. So $3n + 2$ is both odd and even — a contradiction. Hence $p \wedge \neg q$ is false, and therefore $p \to q$ is true. ■

Example 40. Prove that $\sqrt{2}$ is irrational, by contradiction.

Let p be the proposition "$\sqrt{2}$ is irrational".

Proof. Suppose $\neg p$ is true, so $\sqrt{2}$ is rational. Then

$$\sqrt{2} = \frac{a}{b},$$

where a and b are integers with **no common factor** and $b \neq 0$. Squaring both sides,

$$2 = \frac{a^2}{b^2} \quad \text{so} \quad a^2 = 2b^2.$$

Then a^2 is even, and therefore a is even, so $a = 2m$ for some integer m. Substituting,

$$(2m)^2 = 2b^2 \quad \Rightarrow \quad 4m^2 = 2b^2 \quad \Rightarrow \quad b^2 = 2m^2,$$

so b^2 is even and therefore b is even. But now a and b are both even, so they share the common factor 2 — contradicting the assumption that they have none.

Hence $\neg p$ is false, so p is true: $\sqrt{2}$ is irrational. ■

Note. Step "a^2 even $\Rightarrow$ a even" is itself a small theorem, proved by contraposition: if a is odd then a^2 is odd (Section 9.5, Example 36). Naming it as a **lemma** is exactly what lemmas are for.

9.7 Choosing a method

If the claim looks like...	start with
$p \to q$ and p gives you something to compute with	**direct proof** — expand the definition of p
$p \to q$ where q is a negative statement ("not divisible", "not equal")	**contraposition** — $\neg q$ is then positive and usable
an existence claim "there is an x with $P(x)$"	**construct** the witness, or prove it exists non-constructively
a uniqueness claim	prove existence, then assume two witnesses and show they are equal
"no such object exists", irrationality, infinitude	**contradiction** — assume one exists and squeeze out a falsehood
$\forall x P(x)$ over the integers with a base case in sight	**induction** (Chapter 5)
$p \leftrightarrow q$	two proofs: $p \to q$ and $q \to p$

Watch out. Common failures a grader will catch: proving the **converse** instead of the statement; using the conclusion as a hypothesis ("begging the question"); proving an example and calling it a proof; forgetting that the arbitrary element really must be arbitrary.

Exercise set 9

9.1 Give a direct proof that the sum of two even integers is even.

9.2 Give a direct proof that the product of two odd integers is odd.

9.3 Prove by contraposition: if n^2 is even, then n is even.

9.4 Prove by contraposition: if $3n + 2$ is even, then n is even.

9.5 Prove by contradiction: if $n = ab$ where a and b are positive integers, then $a \le \sqrt{n}$ or $b \le \sqrt{n}$.

9.6 Decide which method fits and then prove it: if n is an integer and $n^3 + 5$ is odd, then n is even.

9.7 Find the flaw. *"Theorem: if n^2 is positive then n is positive. Proof: suppose n^2 is positive. We know that if n is positive then n^2 is positive. Therefore n is positive."* Name the error, and give a counterexample to the theorem itself.

10 Quick reference

10.1 Symbols

Symbol	Meaning	Symbol	Meaning
$\neg p$	negation, "not p"	$\forall x$	for all x
$p \wedge q$	conjunction, "and"	$\exists x$	there exists x
$p \vee q$	disjunction, "or"	$\exists! x$	there exists exactly one x
$p \oplus q$	exclusive or	$\equiv$	is logically equivalent to
$p \to q$	conditional, "if-then"	$\therefore$	therefore
$p \leftrightarrow q$	biconditional, "iff"	$\blacksquare$	end of proof

10.2 The three tables you should be able to write from memory

1. The truth table of $\to$ (false only in the row $T \to F$).
2. De Morgan's laws, for connectives **and** for quantifiers.
3. Modus ponens, modus tollens, and hypothetical syllogism.

11 Mixed review problems

These practice problems draw on the whole chapter rather than a single section.

Exercise set 10

10.1 Which of the following are propositions? (a) "$4 + x = 7$"; (b) "Benha is in Egypt"; (c) "Close the door."; (d) "$2^{10} > 1000$".

10.2 Construct the truth table of $(p \to q) \wedge (\neg p \to r)$.

10.3 Write the converse, inverse, and contrapositive of "If it rains, the match is cancelled."

10.4 Translate into logic: "You can graduate only if you pass discrete mathematics and you do not fail the project."

10.5 Show that $\neg(p \to q) \equiv p \wedge \neg q$ using a truth table.

10.6 Show that $(p \wedge q) \to (p \vee q)$ is a tautology.

10.7 Use the laws of logic (not a truth table) to simplify $\neg(\neg p \vee q) \vee (p \wedge \neg q)$.

10.8 Negate, pushing $\neg$ all the way in: $\forall x \exists y (P(x) \wedge \neg Q(x, y))$.

10.9 Show that the premises $p \vee q$, $p \to r$, $q \to r$ imply r.

10.10 Prove: if n is an even integer, then n^2 is even. Which method did you use, and why?

10.11 Prove by contraposition: if n^2 is odd, then n is odd.

10.12 Prove by contradiction: there is no smallest positive rational number.